F13-SW-103
2000-F13-SLD-EL-10911

F13-SR-001
415V SAG MILL SWITCHROOM

SWR EARTH BAR

FEEDER FROM F13-SW-002
2000-F13-SLD-EL-10301

TO F13-SW-002
2000-F13-SLD-EL-10301

F13-SW-102
SAG MILL 415V ESSENTIAL SWITCHBOARD

EARTH BAR

NEUTRAL BAR
NOTE 1

AFD

2A
63A

2A
63A

MANUAL
MCCB CHANGEOVER
SWITCH 3 POLE 630A

1

00F-05

1

415V 3NE~ 50Hz 630A 65kA for 1sec

INC

01F-16
160A
2A
32A
90A
160A
250A

01F-05
32A
2A
32A
70A
32A
32A

01F-21
250A
2A
32A
90A

01F-12
100A
2A
32A
70A
ECR

01F-09
63A
2A
32A
70A

02F-16
250A
2A
32A
90A

02F-11
250A
2A
32A
90A
160A
250A

02F-06
250A
2A
32A
90A
160A
250A

1

69kW
F13-DB-001
SAG MILL
415V SWITCHROOM
L&SP DB
2000-F13-SCM-EL-10305

ISL

10kW
F13UPS001-ISL
240VAC UPS
MAINTENANCE
BYPASS SWITCH
2000-F13-SCM-EL-10306

10kW
F13-DB-201
SAG MILL
415V SWITCHROOM
240VAC UPS DB
2000-F13-TLD-EL-10012

UPS

10kW
F13-UPS-001
SAG MILL
415V SWITCHROOM
240VAC UPS
2000-F13-TLD-EL-10012

69kW
F13-DB-304
MILLING AREA 1
L&SP DB
2000-F13-SCM-EL-10307

CR
48kW
1
F13-CR-402
CLASSIFICATION AREA
OVERHEAD CRANE
2000-F13-SCM-EL-10308

CR
15.9kW
F13-CR-411
MILL LINER
MAINTENANCE HOIST 1
2000-F13-SCM-EL-10309

69kW
F13-DB-307
MILLING AREA 4
L&SP DB
2000-F13-SCM-EL-10364

69kW
F13-DB-308
MILLING AREA 5
L&SP DB
2000-F13-SCM-EL-10973

69kW
F13-DB-003
SAG & BALL MILL
VSD SWITCHROOM
L&SP DB
2000-F13-SCM-EL-10906

NOTES

1. EARTH-NEUTRAL LINK INSTALLED IN UPSTREAM F13-SW-002 INCOMER TIER. SIGNAGE SHALL BE PLACED ON F13-SW-102 SWITCHBOARD READING "M.E.N. LINK LOCATED IN F13-SW-002 INCOMER TIER"
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
4. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.

LEGEND

PFR	PHASE FAILURE RELAY
I >	EARTH LEAKAGE RELAY
xxxA / xxxA	ST TRIP SET POINT / CB SIZE
AFD	ARC FLASH DETECTION
ECR	EARTH CONTINUITY RELAY

NORTHERN STAR

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

2000-F13-SLD-EL-10311_Rev1

F13-SW-102 415V essential switchboard - single line diagram